

Effect of an upstream tall building on a street canyon flow

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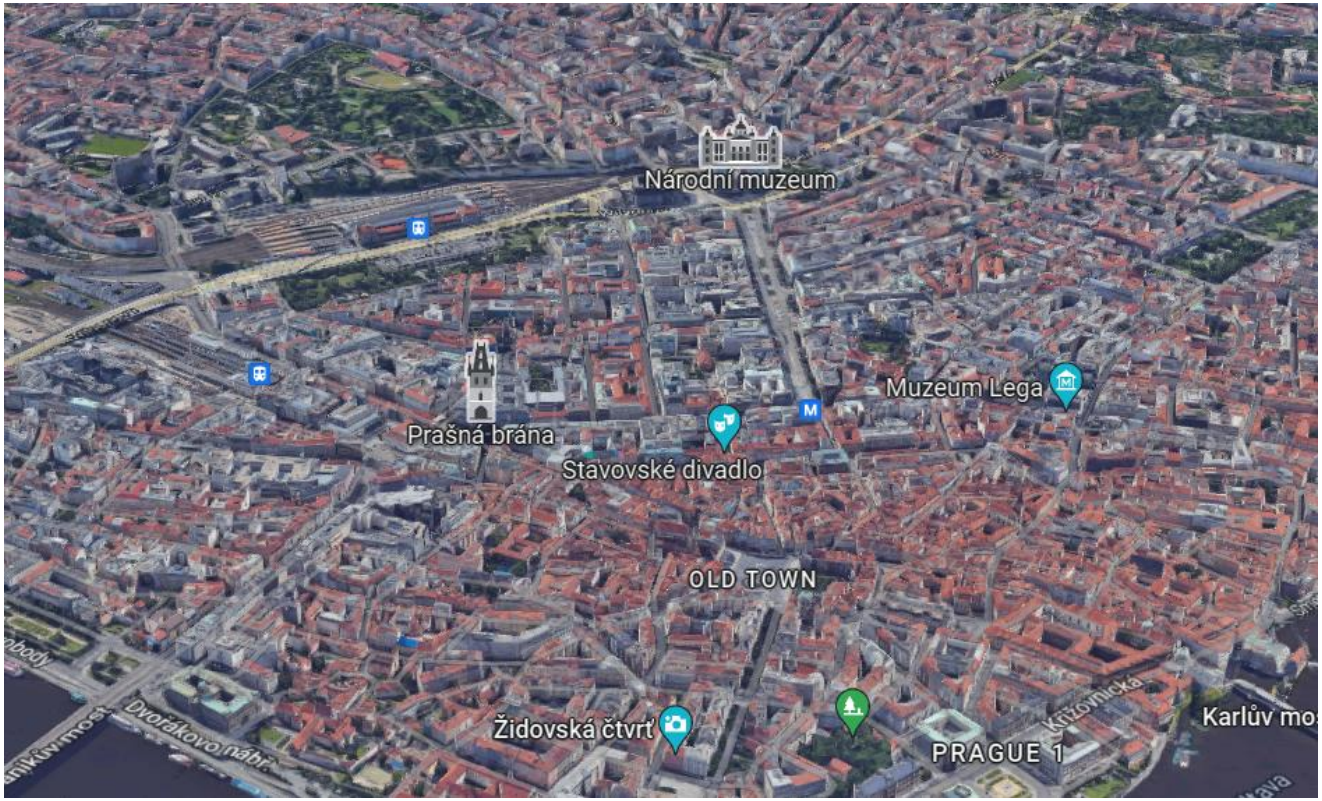
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Overview

- Background
- Objectives
- Experiment set-up
- Results & Discussion
- Conclusions
- Future work

Background >>> General



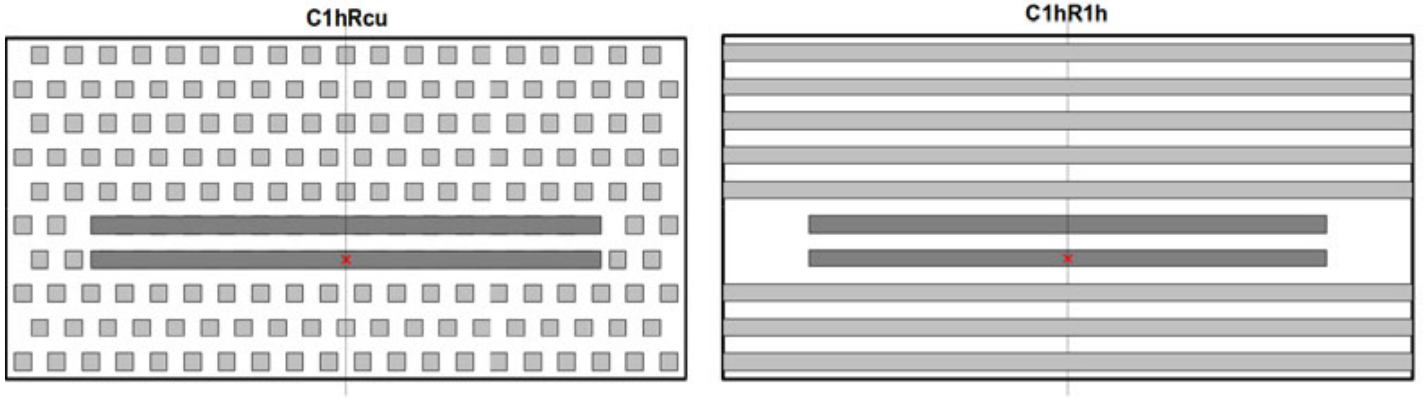
Real urban terrain ← Wind tunnel modeling

Real urban terrain → Idealized model

Real urban terrain → Morphological model

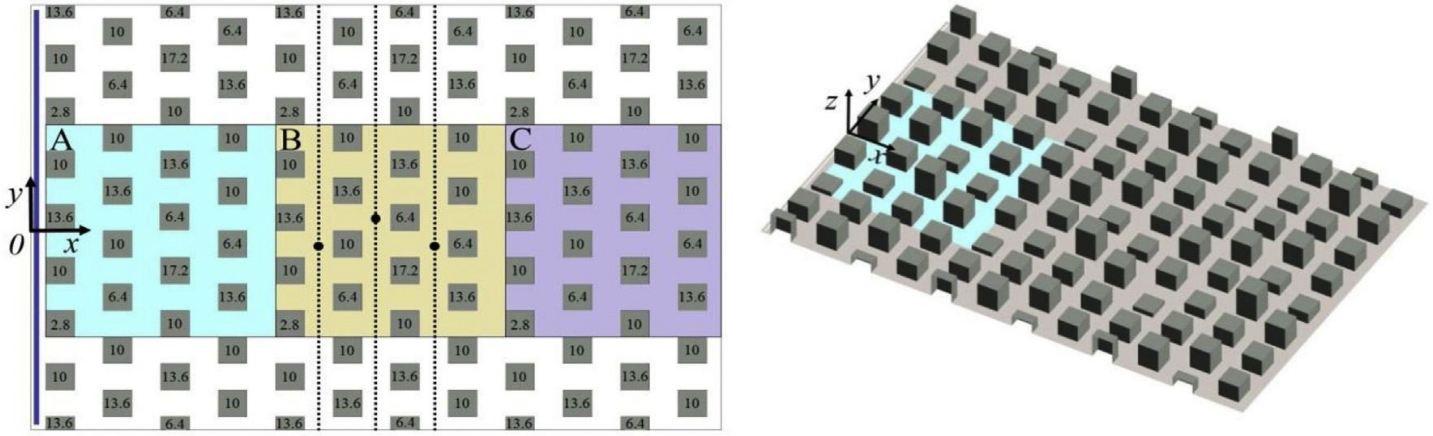
Background >>> Idealized model

Wind direction



Homogeneous model (Blackman et al., 2015)

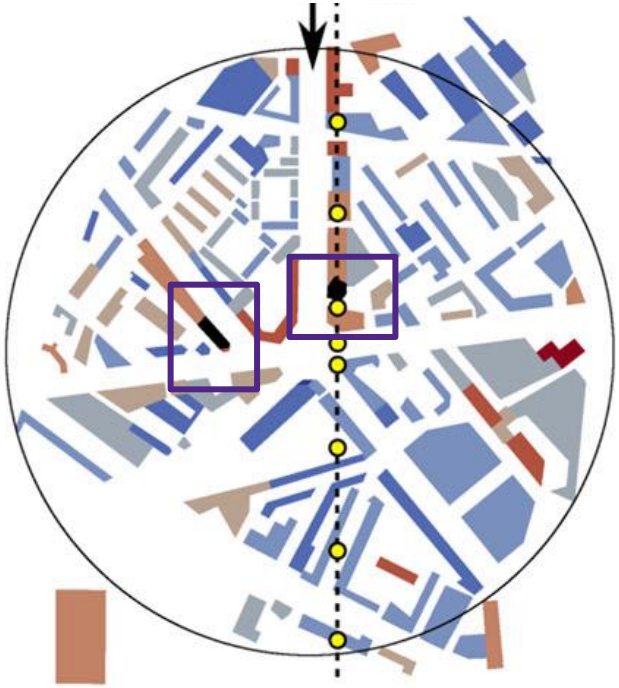
Wind direction

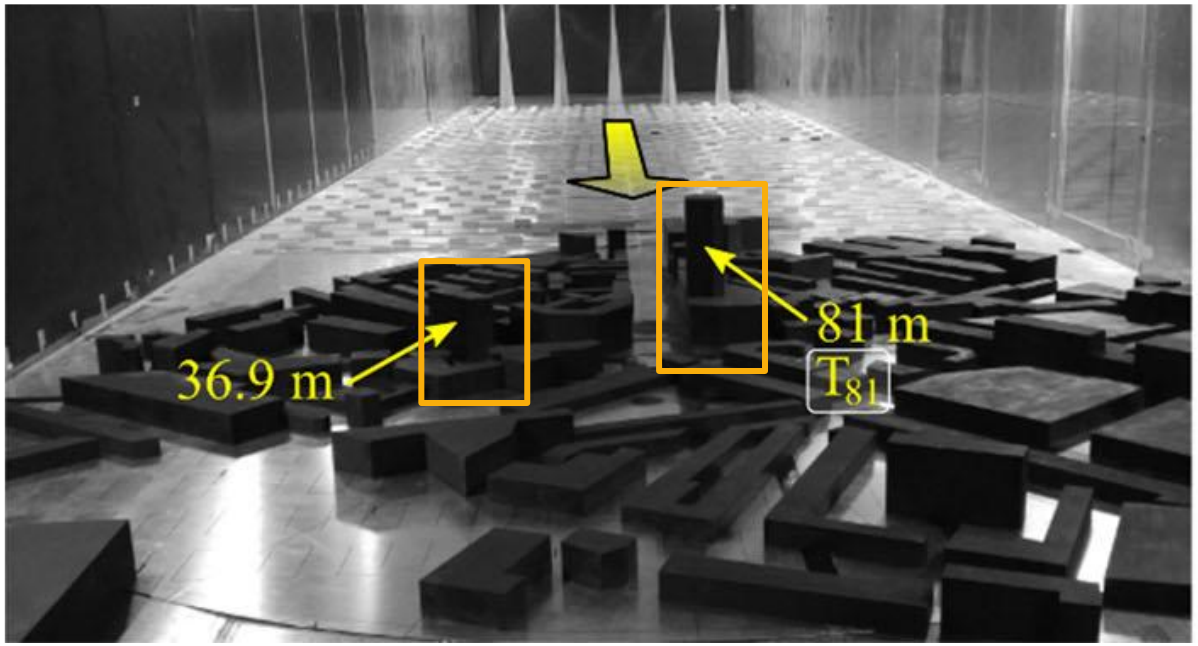


(a) RBSA (b) RBSA

Heterogeneous model (Goulart et al., 2019)

Background >>> Morphological model



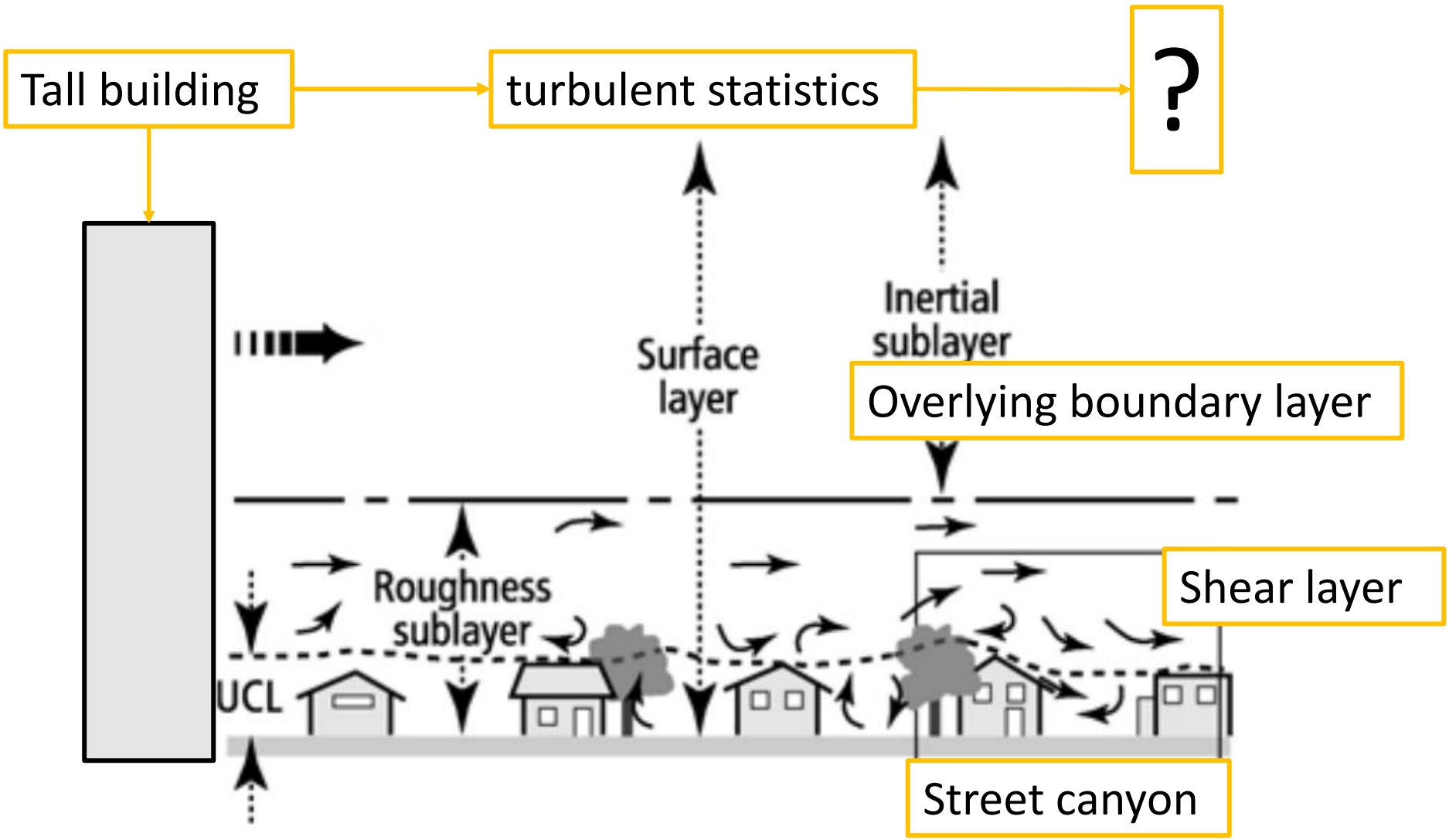


a) Core, b) Core without tall buildings c) Only tall buildings (Hertwig et al., 2019)

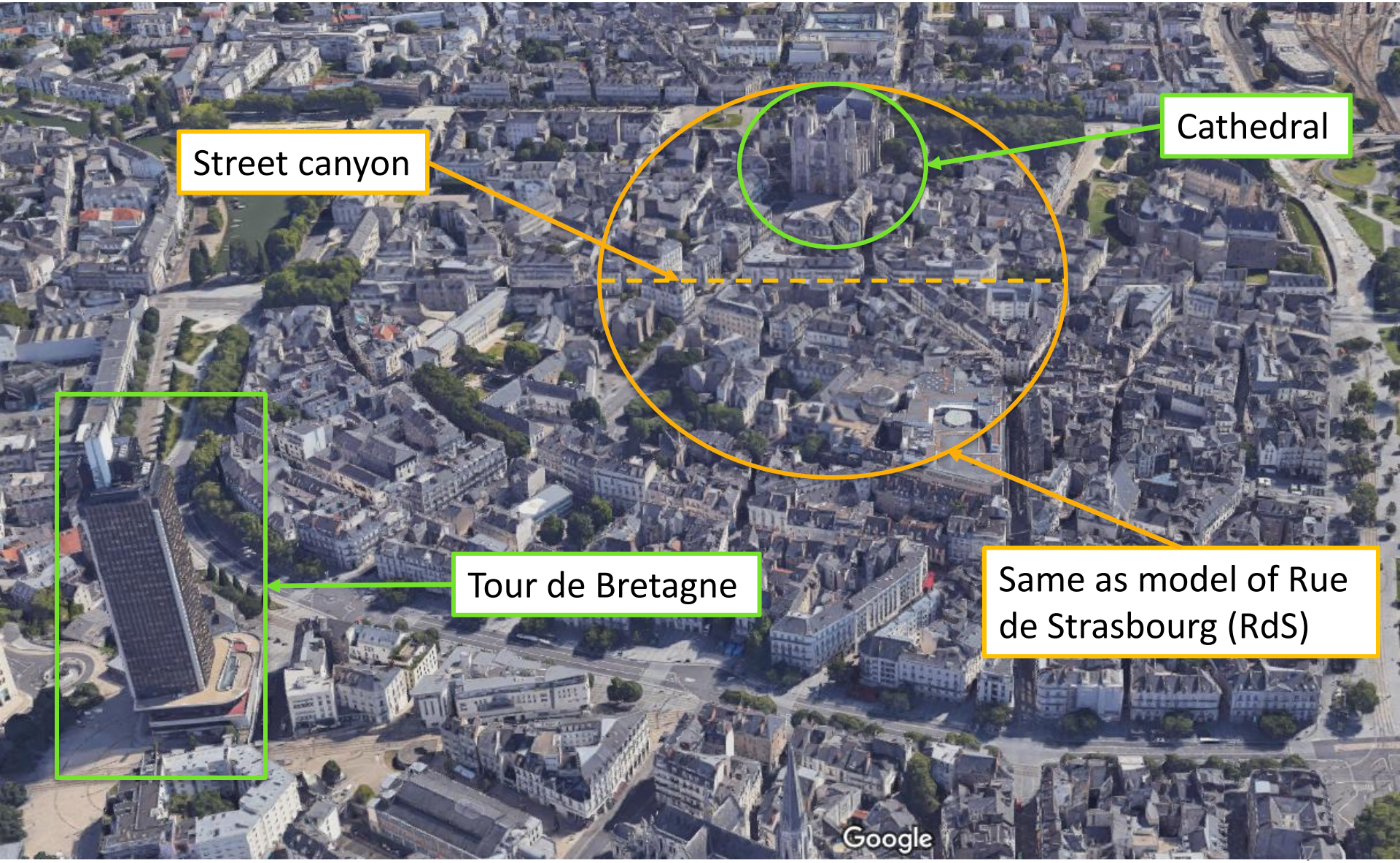
- Recirculation core moves upwards in the near-wake region (a) & (c)
- Lateral velocity fluctuations & turbulence kinetic energy are reduced in the main-wake region (a) & (c)
- Etc.

** $\overline{h_{herwig,no\ tall}}$ = 13.18m/200 = 65.9 mm

Objective

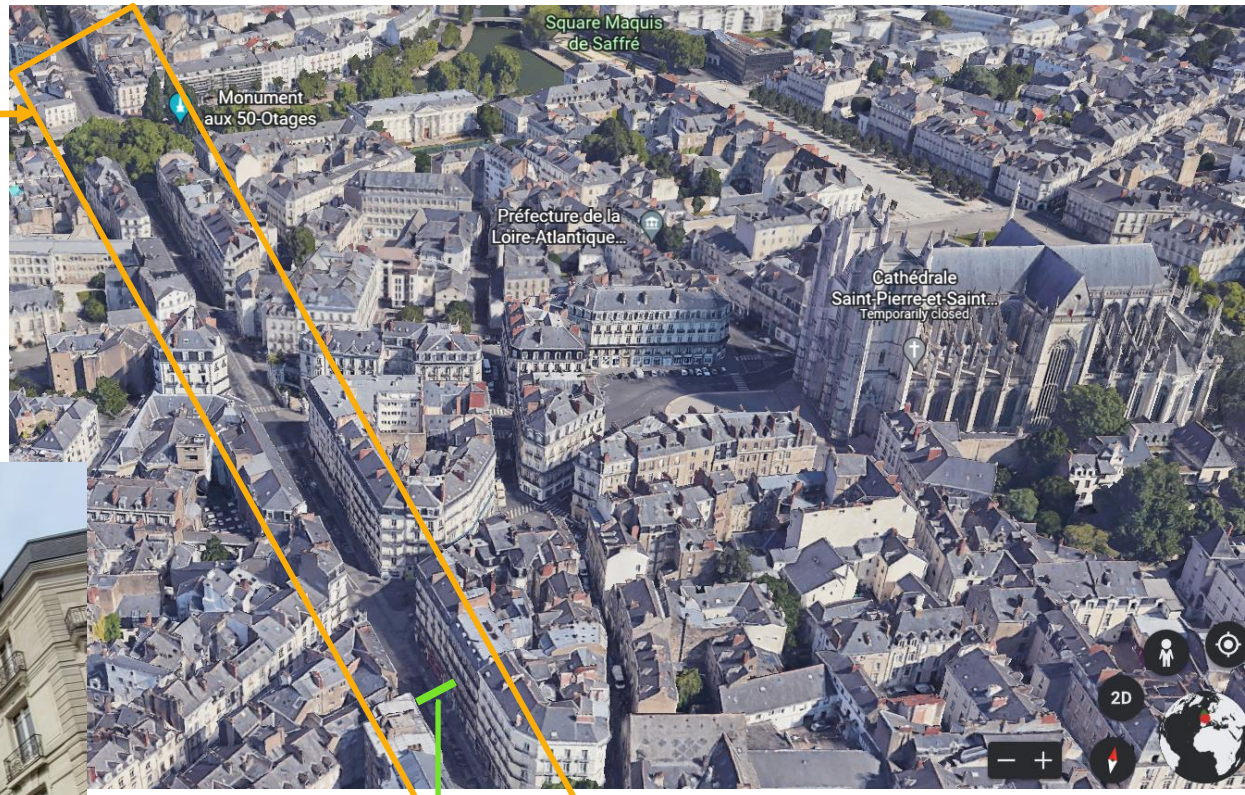

Tall building
turbulent statistics
?
Surface layer
Inertial sublayer
Overlying boundary layer
Roughness sublayer
UCL
Shear layer
Street canyon

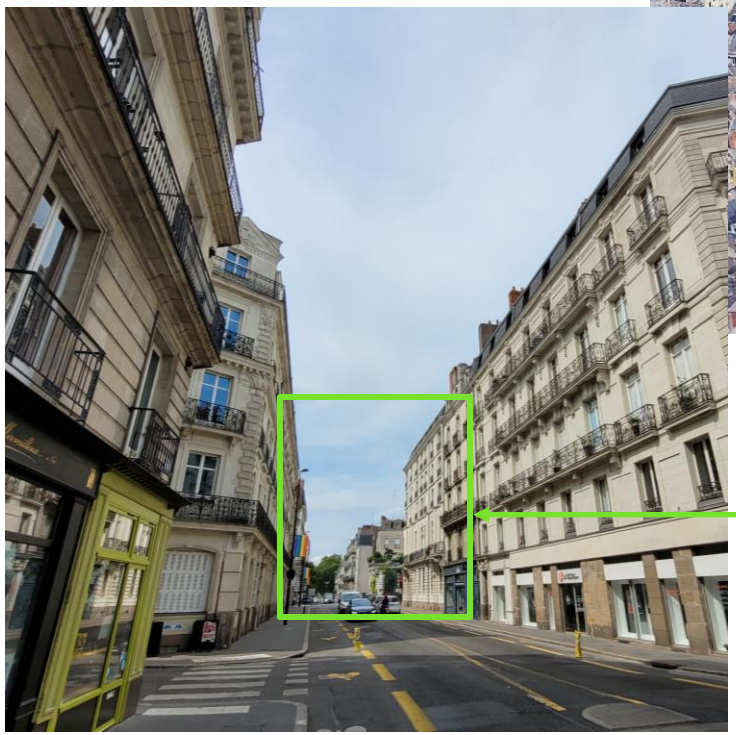
Experiment set-up >>> Model



Experiment set-up >>> Model

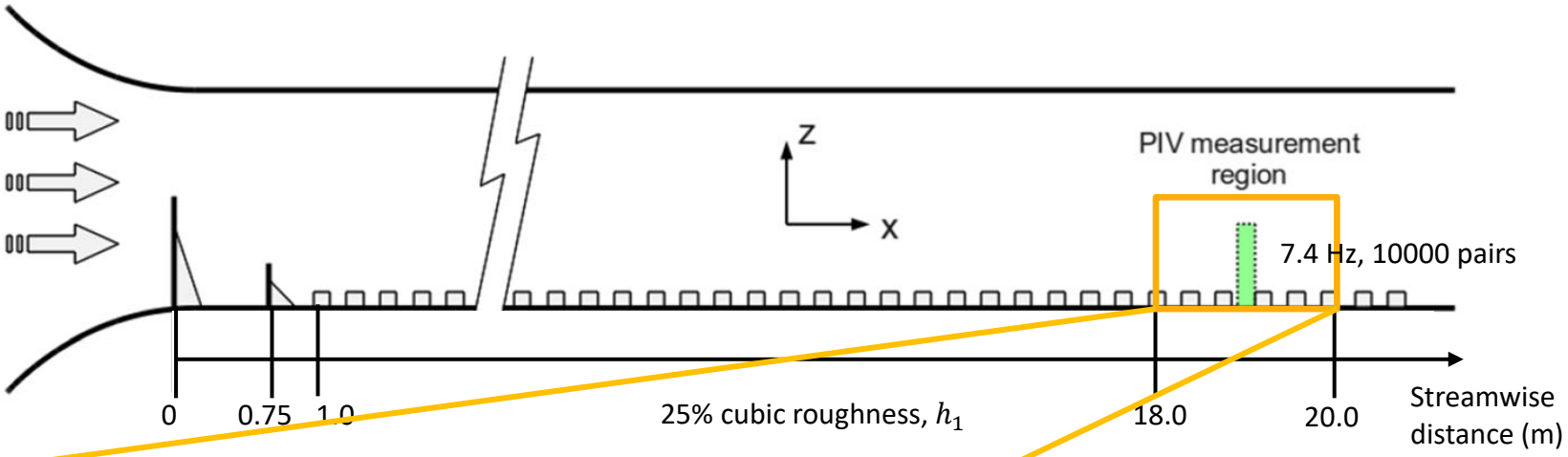
Street canyon

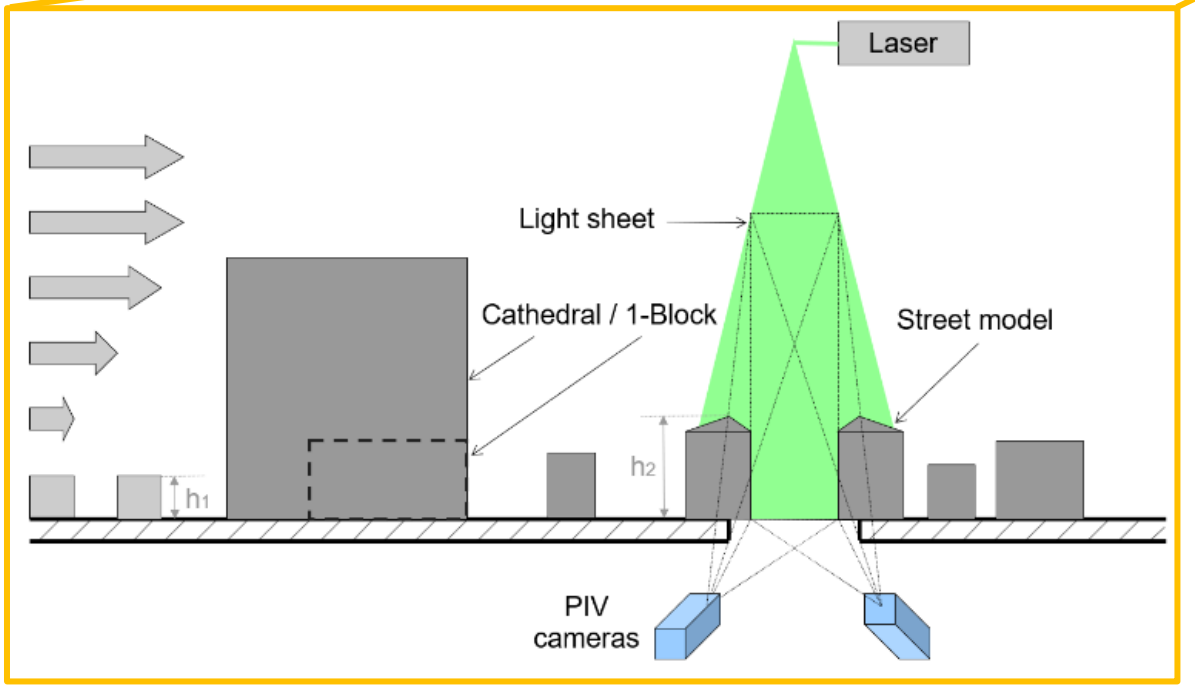




Test plane

Experiment set-up >>> Wind tunnel





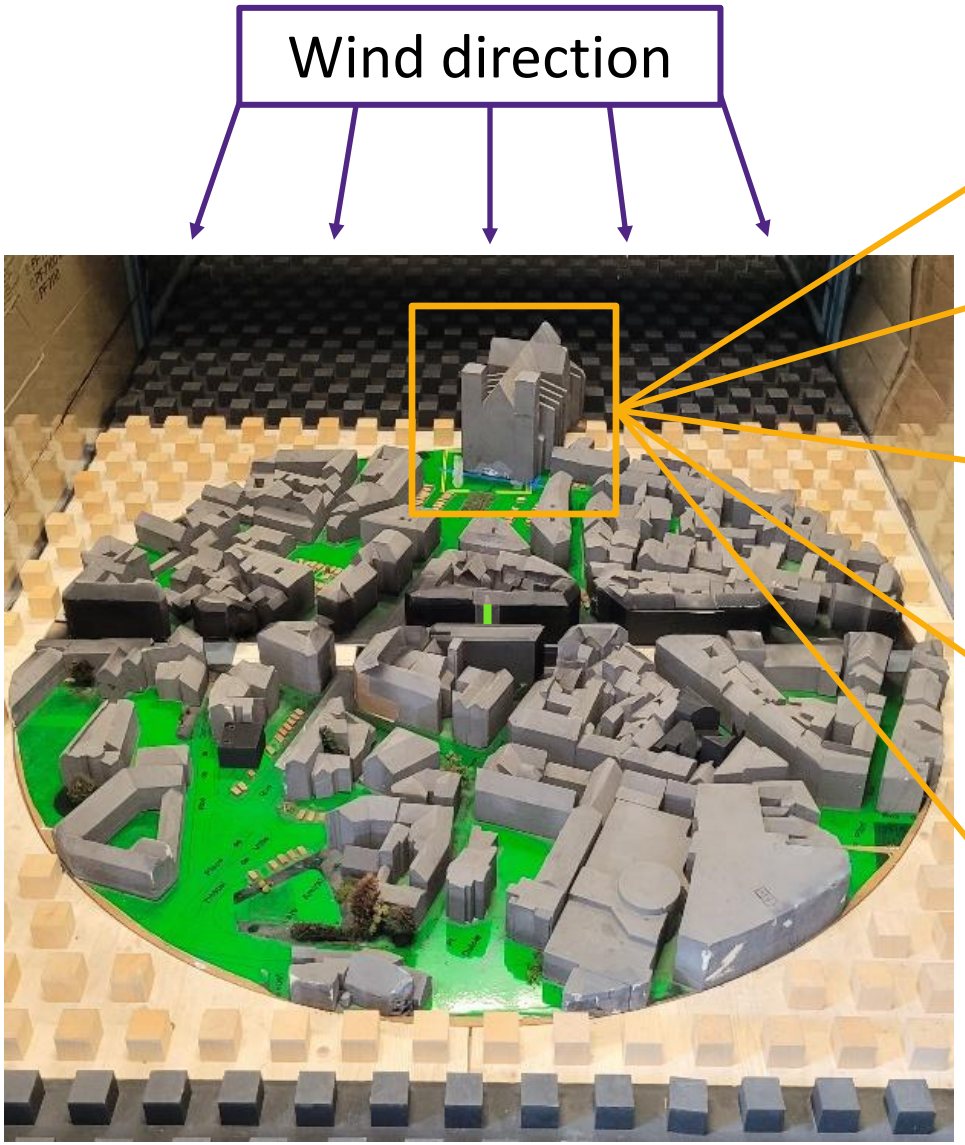
1:200 scaled Model
– Rue de Strasbourg (RdS)

$h_1 = 50\ mm$
– cubic roughness height

$h_2 = 116\ mm$
– street canyon ridge height

$\bar{h} = 100\ mm$
– RdS average height without the cathedral

Experiment set-up >>> Model



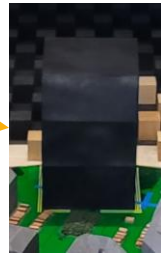
1:200 scaled Model – (RdS)



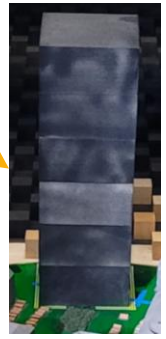
1-block
$2\bar{h}\times2\bar{h}\times1\bar{h}$
90° to the canyon



2-block
$2\bar{h}\times2\bar{h}\times2\bar{h}$
90° to the canyon



3-block
$2\bar{h}\times2\bar{h}\times3\bar{h}$
90° to the canyon



6-block
$2\bar{h}\times2\bar{h}\times6\bar{h}$
90° to the canyon

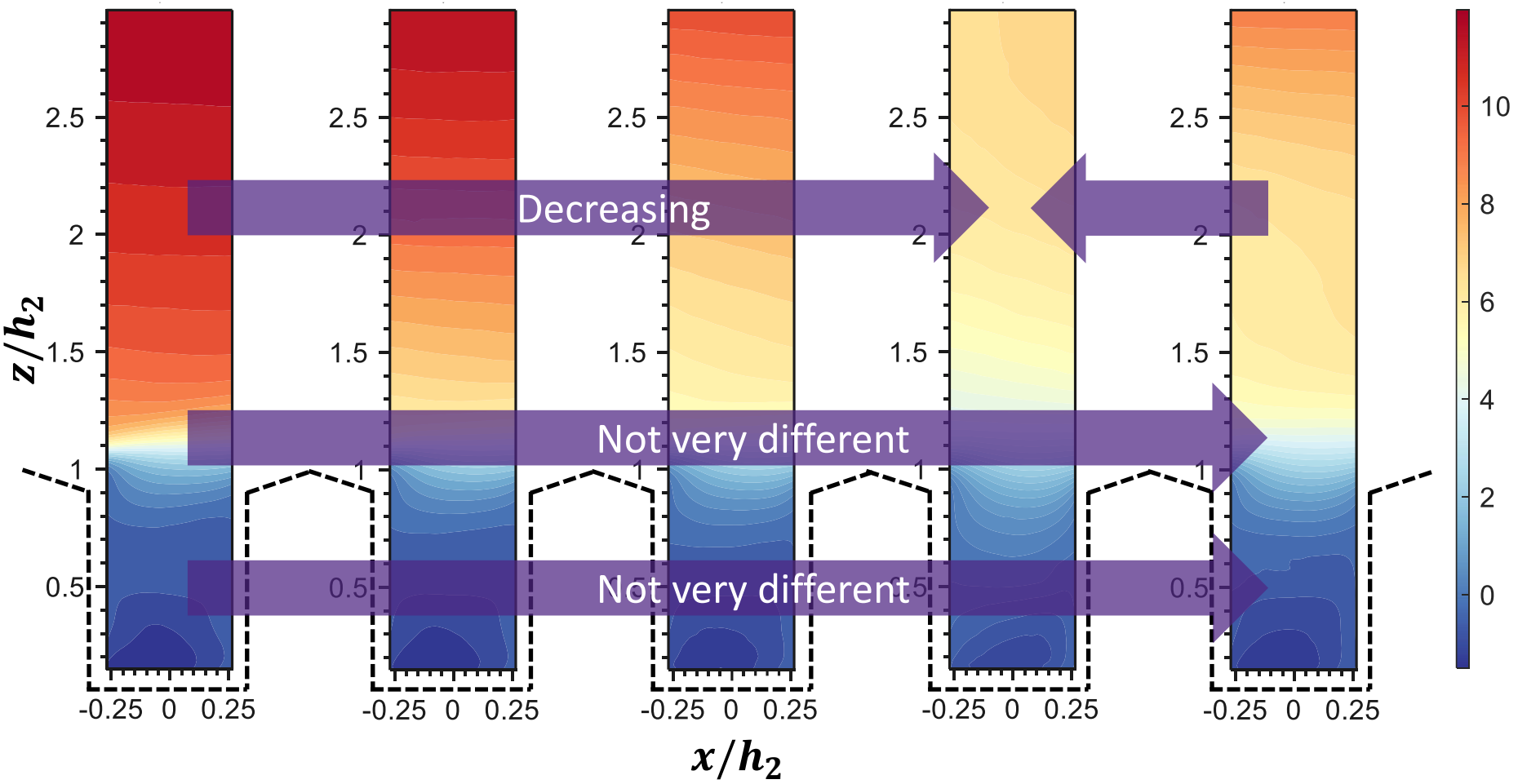


Cathedral
$2\bar{h}\times3.5\bar{h}\times3\bar{h}$
~70° to the canyon

** $\bar{h} = 100\ mm = 0.86h_2$

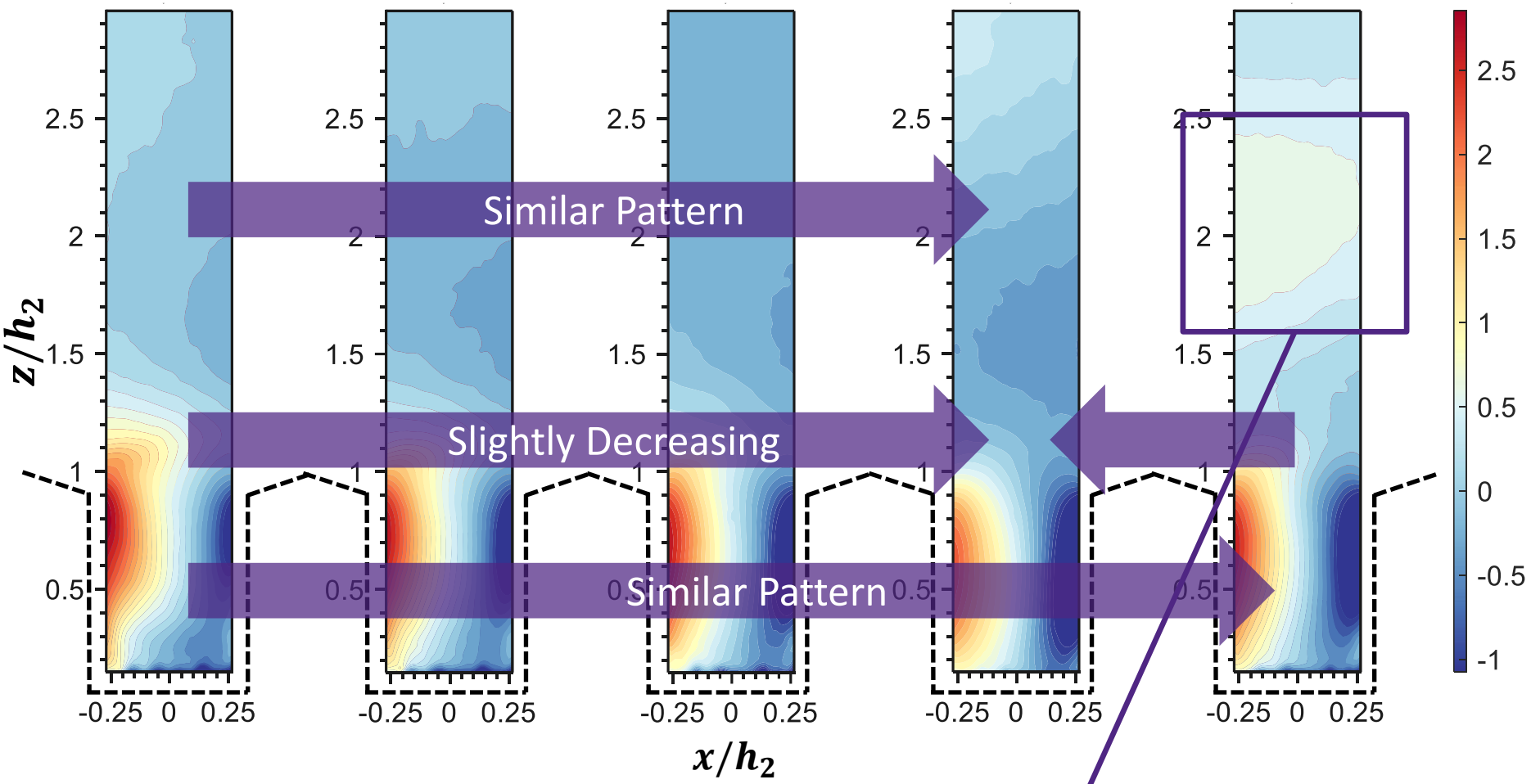
Results & Discussion >>> $\overline{U}/u_*$

** u_* is friction velocity



1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

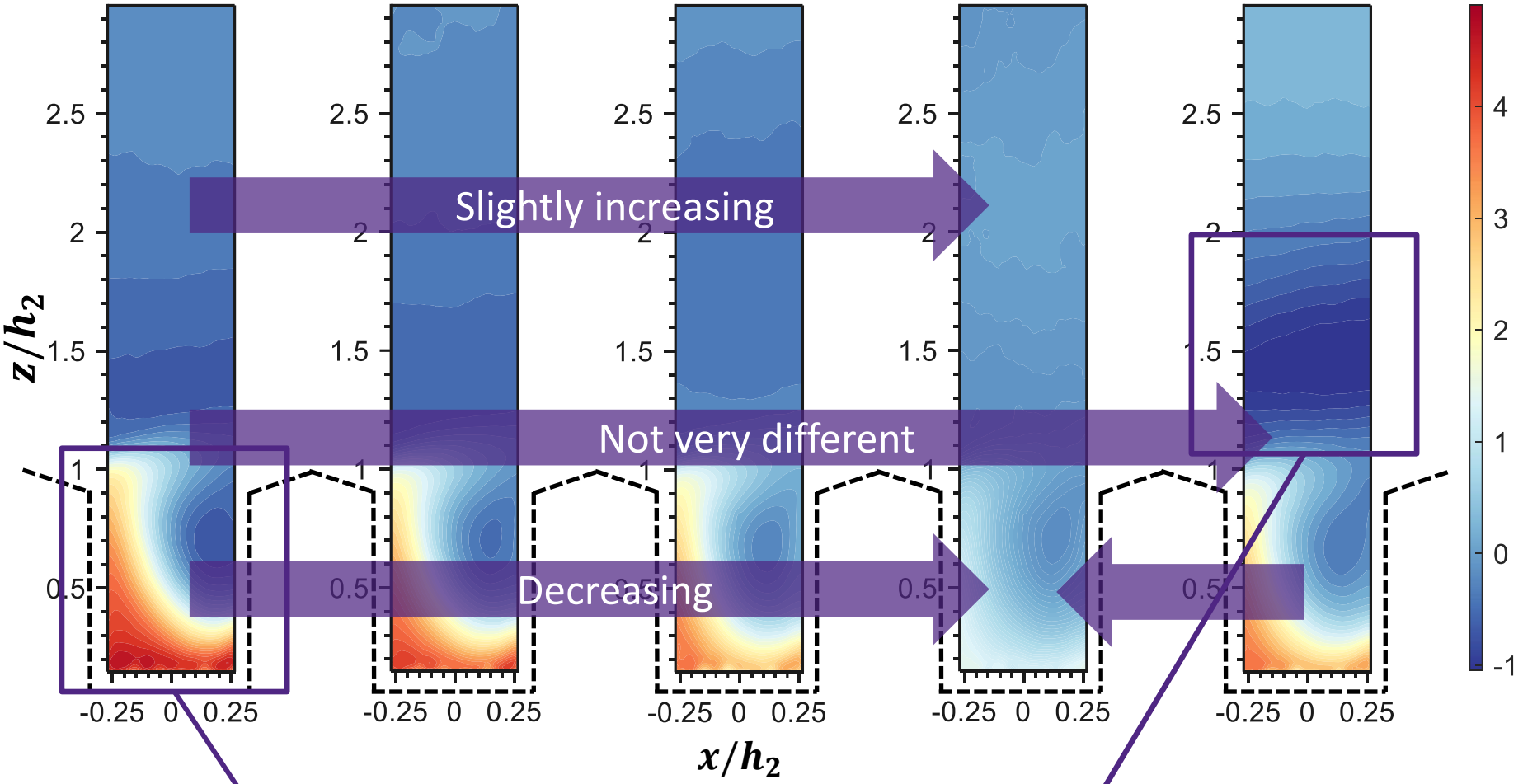
Results & Discussion >>> $\bar{W}/u_*$



1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

Tilted angle & irregular shape of cathedral

Results & Discussion >>> $\bar{V}/u_*$

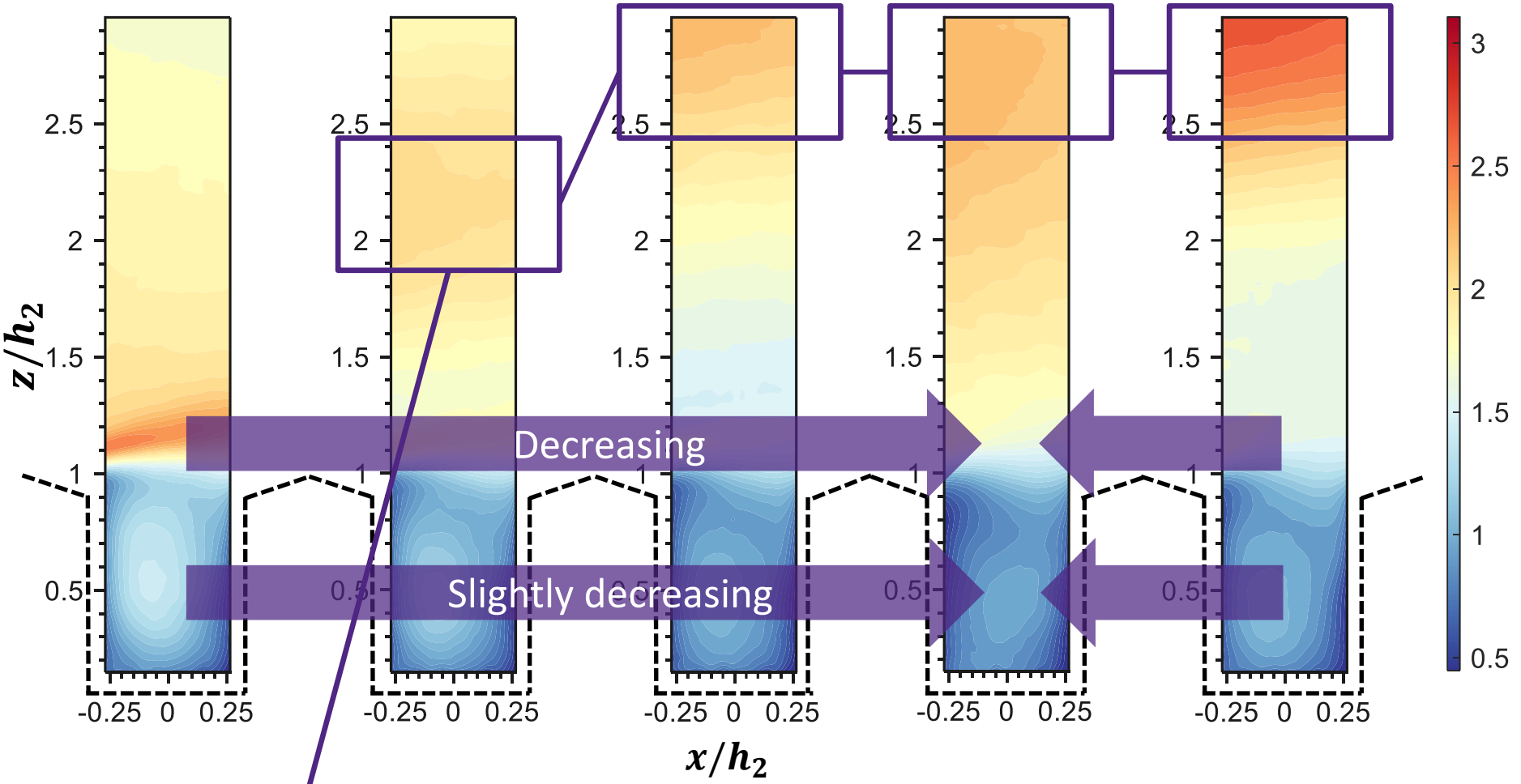


1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

Intersections of street canyon

Tilted angle & Irregular shape of cathedral

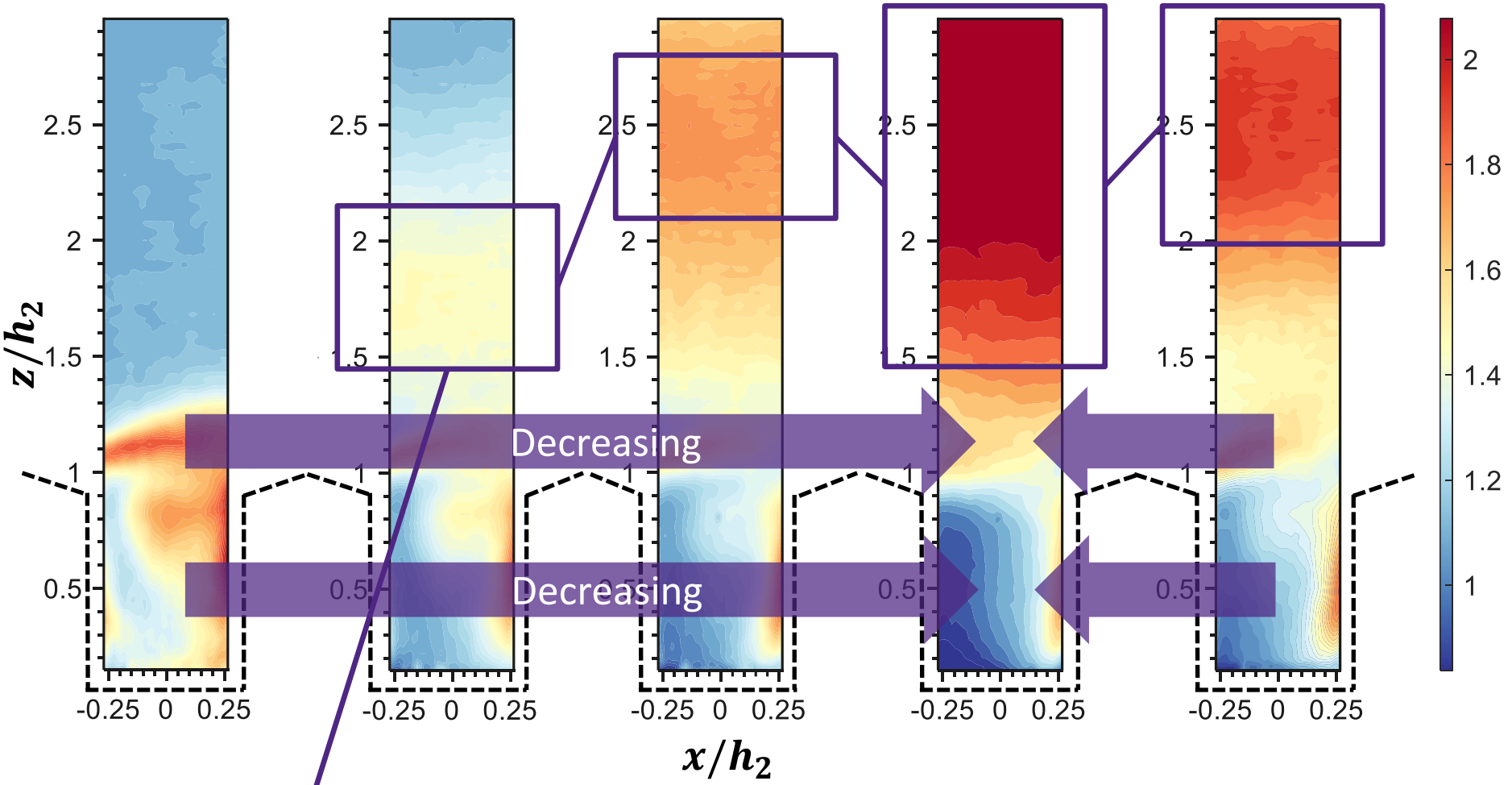
Results & Discussion >>> σ_u/u_*



1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

Upstream tall building

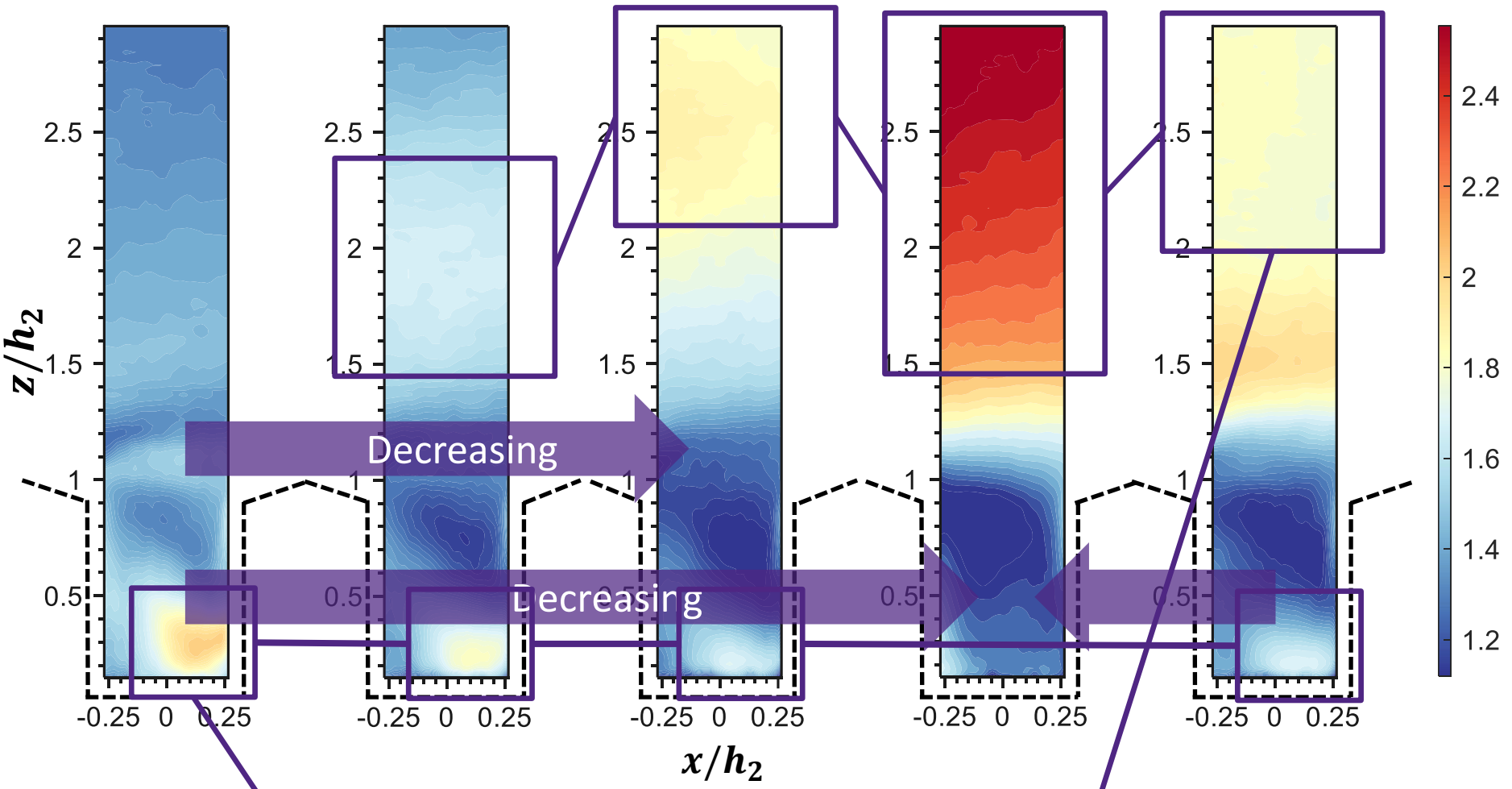
Results & Discussion >>> σ_w/u_*



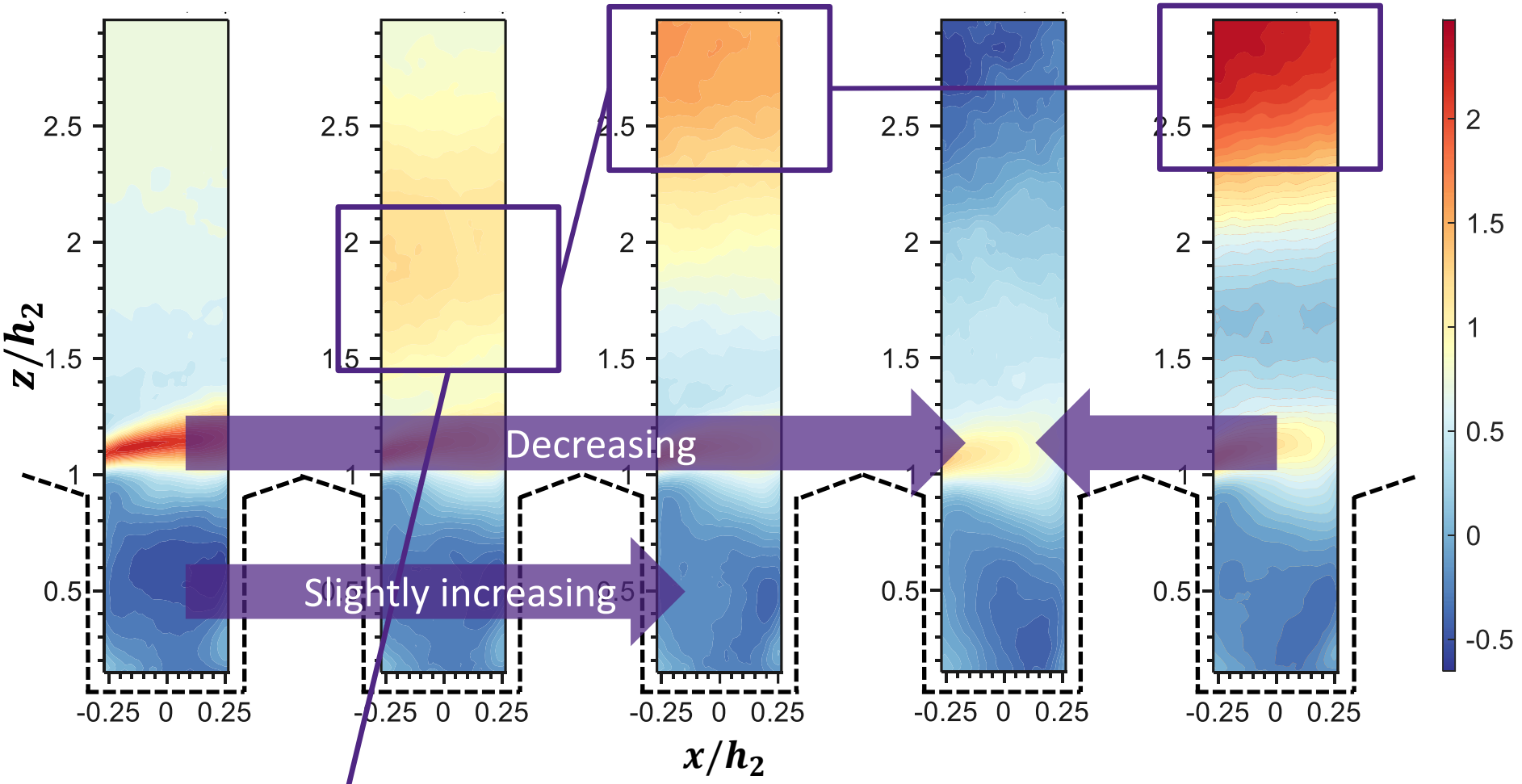
1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

Upstream tall building

Results & Discussion >>> σ_v/u_*



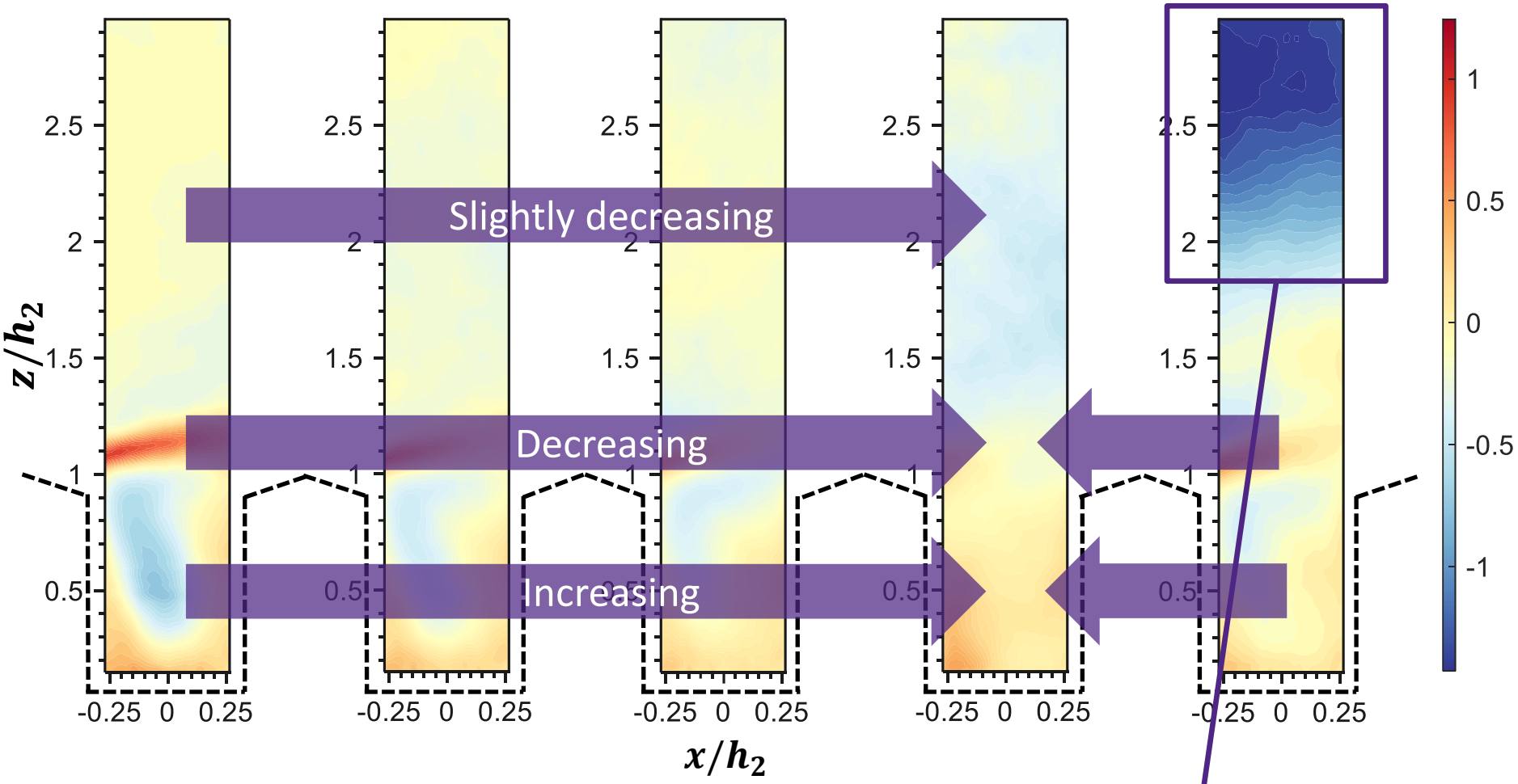
Results & Discussion >>> $-\overline{u'w'}/u_*^2$



1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

Upstream tall building

Results & Discussion >>> $-\overline{u'v'}/u_*^2$

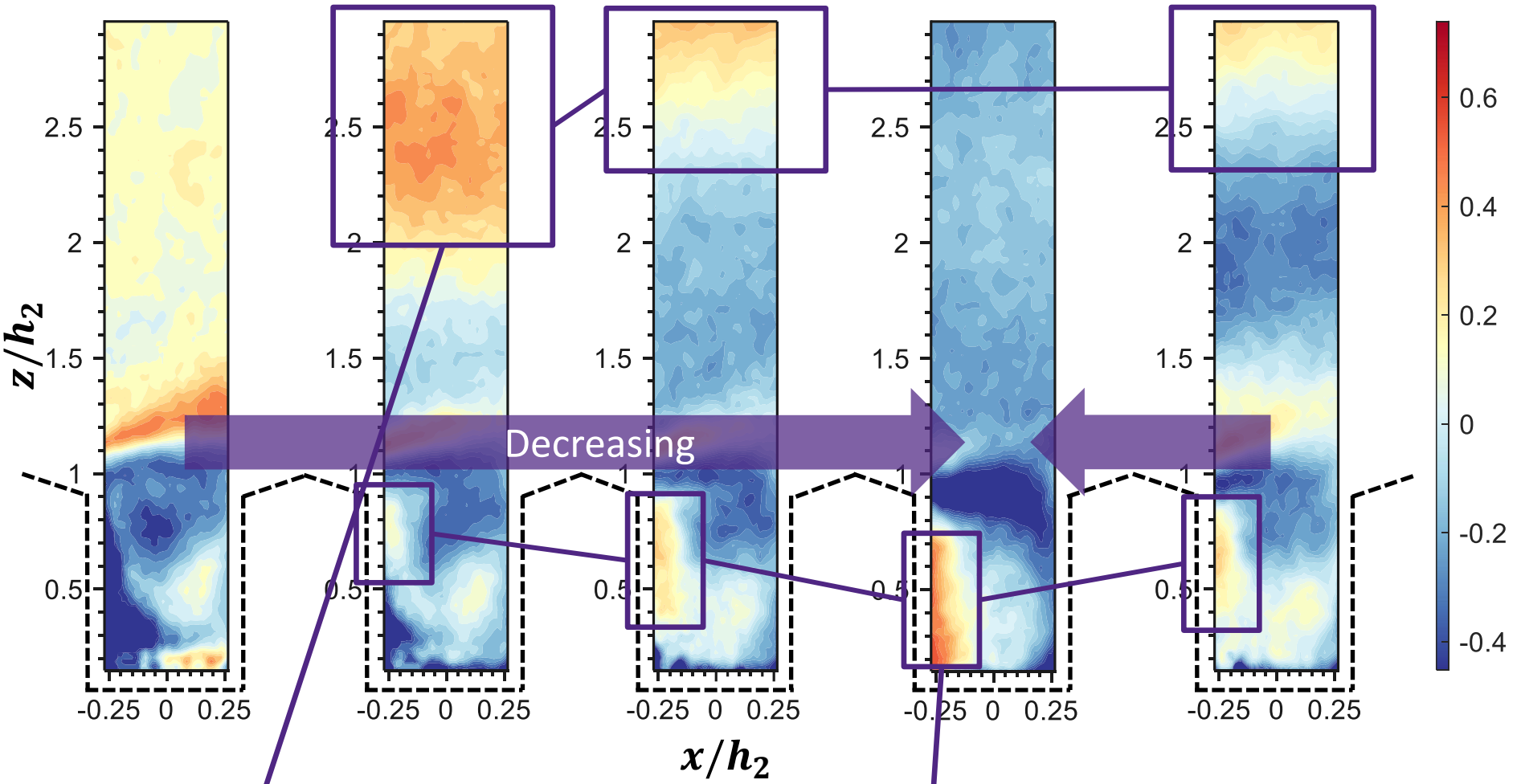


1-block	2-block	3-block	6-block	Cathedral
$0.86h_2$	$1.72h_2$	$2.58h_2$	$5.16h_2$	$2.58h_2$

$-\overline{u'v'} \approx 0$ in idealized config

Shape and direction

Results & Discussion >>> $S_w = \overline{w'^3}/\sigma_w^3$

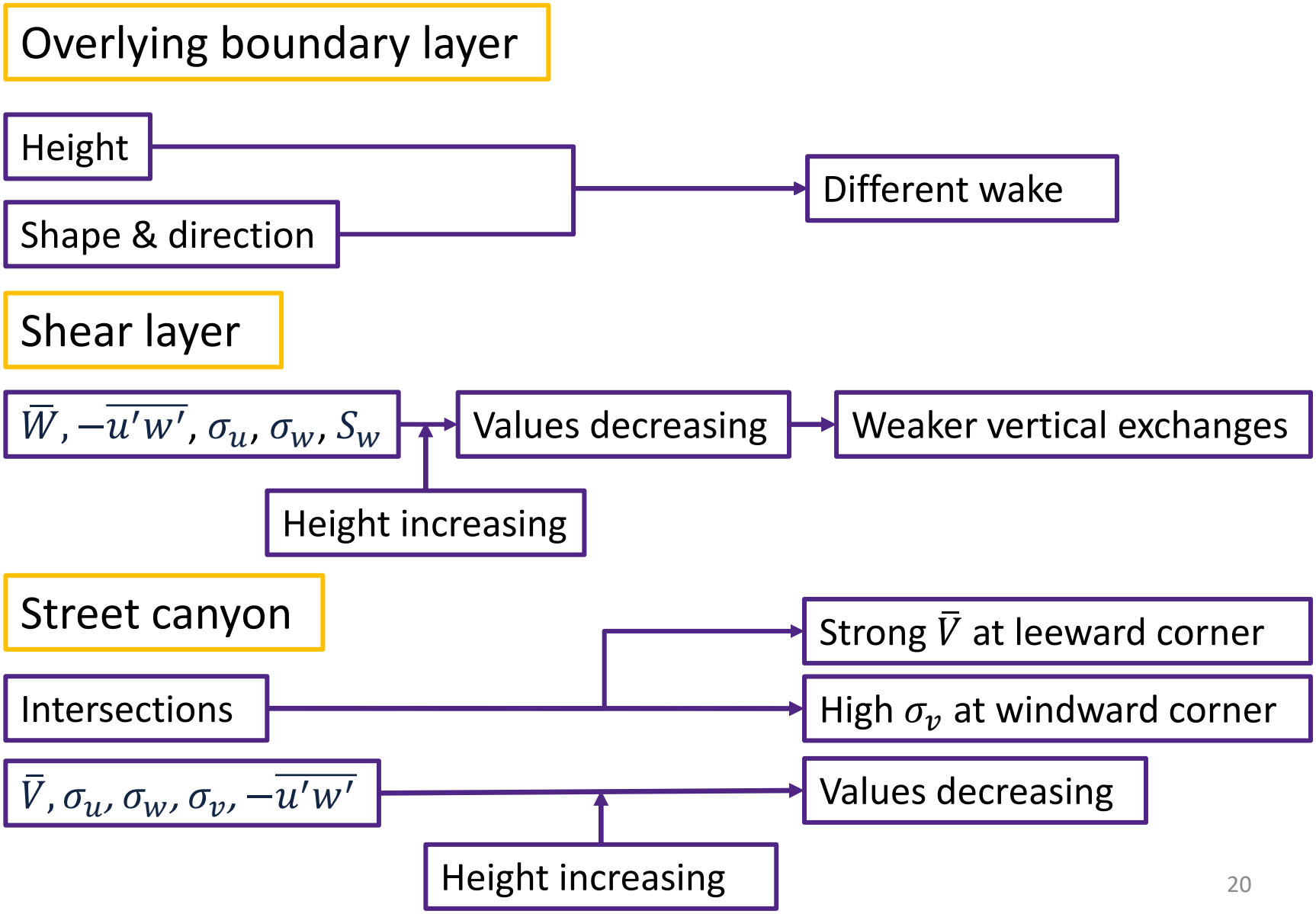


1-block $0.86h_2$ | 2-block $1.72h_2$ | 3-block $2.58h_2$ | 6-block $5.16h_2$ | Cathedral $2.58h_2$

Upstream tall building

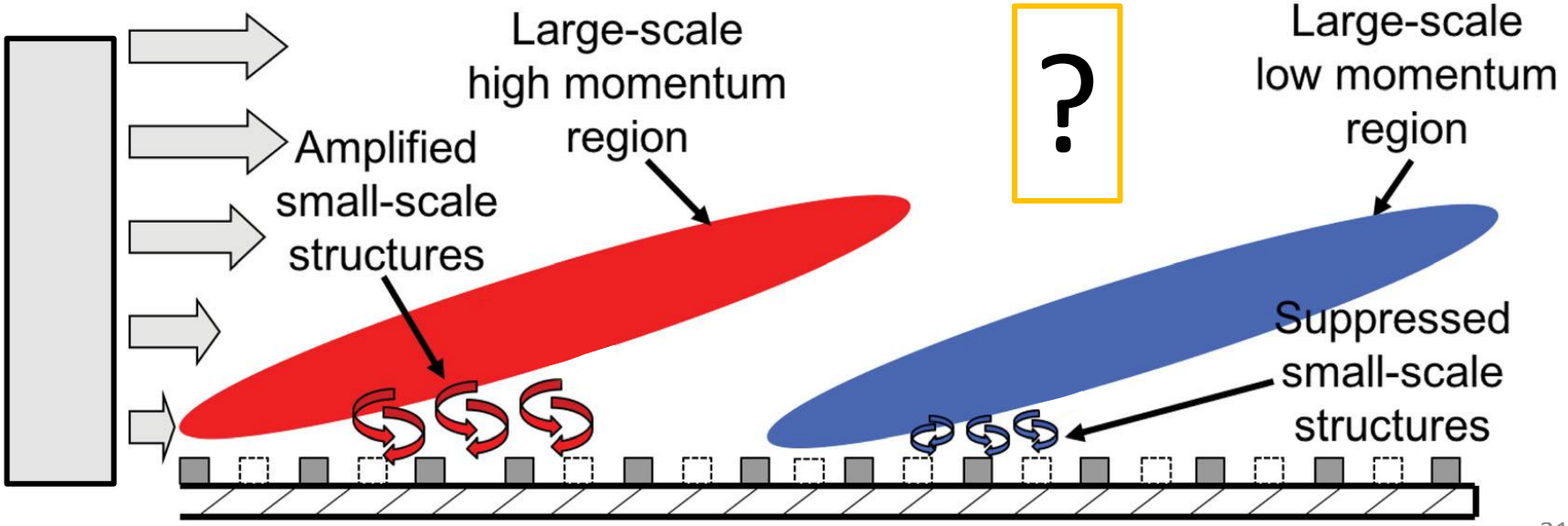
Wind gusts are more probable – pedestrian comfort

Conclusions



Future work

- Compare the turbulent statistics with the idealized street canyon model.
- Characterize the upstream building wake.
- Analyze the interaction between large-scale structures and small-scale structures.
- Analyze the TKE budget.



Acknowledgements





